The search opinion is the **following application documents**

Description, Pages

1-113 as originally filed

Sequence listings, SEQ ID NO

1-40 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/21-21/21 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 Wang Wei ET AL: "Prediction of risk of cardiovascular events in patients with mild to moderate coronary artery lesions using naive Bayesian networks",
Journal of Geriatric Cardiology : JGC, 1 January 2016 (2016-01-01), pages 899-905, XP055928106,
China
DOI: 10.11909/j.issn.1671-5411.2016.11.004
Retrieved from the Internet:
URL:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5253406/pdf/jgc-13-11-899.pdf
[retrieved on 2022-06-03]

D2 US 10 670 611 B2 (SOMALOGIC INC [US]) 2 June 2020 (2020-06-02)

D3 GUANGLIN CUI ET AL: "Polymorphism of tumor necrosis factor alpha (TNF-alpha) gene promoter, circulating TNF-alpha level, and cardiovascular risk factor for ischemic stroke",
JOURNAL OF NEUROINFLAMMATION, BIOMED CENTRAL LTD., LONDON, GB,
vol. 9, no. 1, 10 October 2012 (2012-10-10), page 235, XP021129331,
ISSN: 1742-2094, DOI: 10.1186/1742-2094-9-235

D4 DRAGUT R ET AL: "Relationship between TNF alpha, IL-6 and cardiovascular risk in patients with type 2 diabetes and metabolic syndrome",
ATHEROSCLEROSIS,
vol. 235, no. 2, 2014, XP028878483,
ISSN: 0021-9150, DOI: 10.1016/J.ATHEROSCLEROSIS.2014.05.712

D5 GERSTEIN HERTZEL C. ET AL: "Identifying Novel Biomarkers for Cardiovascular Events or Death in People With Dysglycemia",
CIRCULATION,
vol. 132, no. 24, 15 December 2015 (2015-12-15), pages 2297-2304, XP055928164,
US
ISSN: 0009-7322, DOI: 10.1161/CIRCULATIONAHA.115.015744

D6 FUSTER-PARRA P ET AL: "Bayesian network modeling: A case study of an epidemiologic system analysis of cardiovascular risk",
COMPUTER METHODS AND PROGRAMS IN BIOMEDICINE, ELSEVIER,
AMSTERDAM, NL,
vol. 126, 30 December 2015 (2015-12-30), pages 128-142, XP029433116,
ISSN: 0169-2607, DOI: 10.1016/J.CMPB.2015.12.010

D7 US 2014/255424 A1 (WYSS-CORAY ANTON [US] ET AL) 11 September 2014 (2014-09-11)

D8 EP 2 924 440 A2 (HEALTH DIAGNOSTIC LAB INC [US]) 30 September 2015 (2015-09-30)

D9 EP 3 378 955 A1 (CRITICAL CARE DIAGNOSTICS INC [US]) 26 September 2018 (2018-09-26)

D10 CN 109 797 216 A (UNIV JILIN) 24 May 2019 (2019-05-24)

D11 EP 2 723 893 A1 (GALDERMA RES & DEV [FR]; UNI MUNSTER [DE]) 30 April 2014 (2014-04-30)

D12 ZHAO QIANG ET AL: "Identification of potentially relevant genes for myocardial infarction using RNA sequencing data analysis",
EXPERIMENTAL AND THERAPEUTIC MEDICINE,
vol. 15, 1 January 2018 (2018-01-01), pages 1456-1464, XP093005400,
GR

ISSN: 1792-0981, DOI: 10.3892/etm.2017.5580
Retrieved from the Internet:
URL:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5774508/pdf/etm-15-02-1456.pdf

1 Subject-matter of the application

1.1 The present application relates to methods for the identification of individuals who are at risk from suffering from a cardiovascular event (cardiovascular death, myocardial infarction, stroke or heart failure). The application is based on the analysis of previous patient data using computational analyses to build a disease network representative of cardiovascular disease (Bayesian network analyses). The application refers to specific protein biomarkers and SNP markers for determining, diagnosing and/or prognosing an individuals risk of suffering from a cardiovascular event.

2 Unity of invention - Art. 82 EPC

2.1 The present application lacks unity within the meaning of Article 82 EPC for the following reasons:

Independent claim 1 refers to a method of determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease, wherein in a sample the expression level, amount and/or activity of two or more biomarkers selected from the list given in claim 1 is detected. **Independent claim 9** refers to a method of determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease, wherein in a sample a single nucleotide polymorphism (SNP) in the RORA gene is detected. **Independent claim 11** refers to a method of determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease, wherein in a sample a single nucleotide polymorphism (SNP) in the GHR gene is detected.

The common matter linking the independent method claims can be seen as the detection of one or more marker(s) in the context of determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease.

This common matter however does not comprise a single general inventive concept because the common matter as defined above is already known from the prior art.

Document D1 refers to the establishment of a risk prediction model that can evaluate a patient with certain risk factors for the likelihood of the occurrence of coronary heart disease event within one year. The Risk Estimation Model and the Simplified Model were conducted using Bayesian networks and compared with Single Factor Models. D1 identifies a significant correlation between cardiovascular events and the biomarkers OPG, PIGF, Cathepsin S, GM-CSF, IP-10, CXCL-16, MIP1b, LDL-C, HDL-C, BMI, morphology of coronary lesions, creatine and high blood pressure, see D1 the whole document, especially the abstract and section "discussion" on pages 903-904. Equally **document D2** refers to the detection of at least two biomarkers for evaluating the risk of a future cardiovascular event in an individual over a 1 to 5 year period, wherein the events include myocardial infarction, stroke, congestive heart failure or death. D2 teaches that the likelihood of the subject having a cardiovascular event within 4 years is high if the level of at least ErbB3 is higher than a control level of ErbB3, and if the level of GDF11 is lower than a control level of GDF11, see D2 abstract and the claims. Furthermore **document D3** refers a study investigating the association between common variants in TNF-alpha gene and stroke. The document for example identifies two single nucleotide polymorphisms (SNPs), rs1800629 and rs1799964, to be associated with significant risk of stroke, see D3 abstract; page 4, left column, 3rd paragraph - page 9, left column, 1st par.

Thus the common matter linking together the claims of the present application, is already known from the prior art, as each of D1, D2 and D3 disclose biomarker(s) for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease/event.

The technical features, representing the difference over the non-inventive common matter are the different biomarkers applied in the claimed methods for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease. However, there is no technical relationship between the remaining technical features, because the remaining technical features, i.e. the different biomarkers (protein biomarkers or SNPs), are not of similar nature or share common molecular structures.

The present application may therefore be considered to refer to the problem of providing a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease, providing a corresponding kit, and use of some of the claimed biomarkers in diagnosis/ prognosis, wherein the application provides a plurality of alternative solutions to

this problem by providing a plurality of different markers for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease.

The present application therefore refers to several inventions, not so linked as to form a single general inventive concept. The number of inventions is "n", wherein the "n" is the number of different biomarkers ("at least two biomarkers" or a SNP) as disclosed in claims 1, 9 and 11. The first invention concerns the first biomarker mentioned in claim , i.e. the TNF and GSTA1 alone or in combination with other markers as disclosed in claim 1, and the last invention (the nth invention) concerns the last biomarker mentioned in claim 11, i.e. a SNP in the GHR gene.

Due to the large number of inventions, the inventions listed here are summarized and only five inventions are listed. It is noted that within invention 3 (invention "n-2") multiple inventions, each representing a different "at least two markers alone or in combination with another marker(s) selected from the markers as disclosed in claim 1", are disclosed.

Consequently the present application is considered to refer to the following separate inventions:

Invention 1: claims 1, 2, 5-8 (all partially), a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease by detecting in a sample the expression level, amount and/or activity of TNF and GSTA1 alone or in combination with other markers selected from the markers as disclosed in claim 1, and a corresponding kit.

Invention 2: claims 1, 2, 5-8 (all partially), a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease by detecting in a sample the expression level, amount and/or activity of NT-proBNP and RORA alone or in combination with other markers selected from the markers as disclosed in claim 1, and a corresponding kit.

Invention 3 (invention "n-2"): claims 1-8 (all partially), a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease by detecting in a sample the expression level, amount and/or activity of two or more biomarkers selected from the biomarkers as disclosed in claim 1, and and a corresponding kit.

Invention 4 (invention "n-1"): claims 9, 10 (completely), 13-15 (partially), a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease by detecting in a sample a single nucleotide polymorphism (SNP) in the RORA gene, and medical use of the RORA.

Invention 5 (invention "n"): claims 11, 12 (completely), 13-15 (partially), a method for determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease by detecting in a sample a single nucleotide polymorphism (SNP) in the GHR gene, and medical use of the GHR.

As the claims do not comprise the same or corresponding special technical features, they lack the technical relationship between their subject-matter required by Rule 44 EPC and the claims are not so linked as to form a single general inventive concept.

The application therefore does not fulfil the requirement for unity of invention within the meaning of Article 82 EPC.

2.2 In response to the invitation to pay additional search fees, the Applicant indicated in his letter dated 7.10.2022 to have two further inventions searched, namely two marker combinations falling within the "invention 3" as grouped above. However, the Applicant requested the search of i) IL-6 in combination with any other biomarkers as listed, and the search of ii) RORA in combination with any other biomarkers as listed and argued that each of i) and ii) would represent unified subject-matter. Alternatively the Applicant requested the search of IL-6 in combination with GHR and the search of RORA in combination of TNF in case i) and ii) are considered to cover a plurality of inventions.

IL-6 for the second invention to be searched and and RORA for the 3rd invention to be searched cannot however be considered respectively as the unifying feature within the respective inventions. Both of these markers are disclosed in the prior art in the context of evaluating the risk of a future cardiovascular event in an individual. For example **document D4** refers to a study evaluating cardiovascular risk in patients with metabolic syndrome (MetS) and type 2 diabetes calculated using UKPDS software. D4 discloses the detection of plasma biomarkers and shows that increased plasma levels of pro-inflammatory cytokines TNF alpha and IL-6 are associated with increased cardiovascular risk, see D4 abstract "EAS-0943". Furthermore **document D10** refers to the use of RORA gene in predicting risk of acute myocardial infarction (AMI) in patients with coronary heart disease. D10 discloses that the relative

expression of RORA mRNA in peripheral blood of patients with AMI was significantly higher than that of SCAD (stable coronary artery disease) patients , see D10, page 1, section "technical field", section 2.3, and the claims.

2.3 Therefore the specific marker combinations to be searched are "interleukin 6 (IL-6) in combination with growth hormone receptor (GHR)", numbered now as 2nd invention searched, and "retinoid acid receptor-related orphan receptor alpha (RORA) in combination with TNF", numbered now as 3rd invention searched as requested by the Applicant. This search opinion is thus given to three separate inventions referring to the marker combinations TNF and GSTA1 alone or in combination with other markers selected from the markers as disclosed in claim 1 (1st invention searched; claims 1, 2, 5-8 (all partially)); IL-6 and GHR alone or in combination with other markers selected from the markers as disclosed in claim 1 (2nd invention searched; claims 1, 3, 5-8 (all partially)), and RORA and TNF alone or in combination with other markers selected from the markers as disclosed in claim 1 (3rd invention searched; claims 1, 2, 5-8 (all partially)).

3 Novelty and Inventive step / 1st Invention searched (TNF+GSTA1)- Art. 54 and 56 EPC

3.1 **Document D7** refers to methods of diagnosis, prognosis and monitoring of altered neural cell regenerative capacity and/or altered cognitive function using biomarkers that have been linked to biological aging process. Such biomarkers include for example GSTA1 and TNF alpha. Further the document refers to kits for carrying out the disclosed methods, the kits comprising reagents specific to the biomarkers. Further the kits may also comprise a set of reference values for control population, see D7 par. [0005], [0103], [0160]-[0171]. Thus due to the disclosure of D7, the subject-matter of claim 8 is not novel, Art. 54 EPC.

It is noted that the intended use of the claimed kit ("for determining, diagnosing and/or prognosing the risk of an individual suffering from cardiovascular disease", "the kit is used to identify a decrease in expression..") is not a restrictive feature to the claimed product. The kits of D7 comprising the same components as the kit claimed by the present application can equally be considered to be suitable for the use as defined in claim 8.

3.2 **Document D3** discloses that patients with acute ischemic stroke compared to controls without stroke have significantly higher median serum levels of TNF-alpha. TNF-alpha levels were also positively associated with some established

traditional cardiovascular risk factors, age, BMI, total cholesterol, blood pressure and inversely with HDL, see D3 page 9, left-column, 2nd par - right column, 2nd par.

Document D4 refers to a study evaluating cardiovascular risk in patients with metabolic syndrome (MetS) and type 2 diabetes calculated using UKPDS software. D4 discloses the detection of plasma biomarkers including TNF alpha, and shows that increased plasma levels of pro-inflammatory cytokines TNF alpha and IL-6 are associated with increased cardiovascular risk, see D4 abstract "EAS-0943".

Document D5 refers to the analysis of 237 biomarkers in blood samples. Forward-selection Cox regression models were used to identify biomarkers that were each independent determinants of 3 different incident outcomes: (1) the composite of myocardial infarction, stroke, or CV death; (2) these plus heart failure hospitalization or revascularization; and (3) all-cause death. The analysed biomarkers include for example glutathione S transferase alpha, tumor necrosis factor alpha, NT-proBNP and TFF3. D5 identifies glutathione S transferase alpha level in blood samples to be inversely associated with all three outcomes, see D5 abstract; page 2298; page 2302, right-column; Supplemental material pages 2-4.

Taking D3 or D4 as the closest prior art teaching elevated TNF alpha levels to be associated with cardiovascular risk, the objective technical problem can be seen as the provision of an alternative/improved method of determining cardiovascular risk. The application of the marker glutathione S transferase alpha as an additional marker in the risk assessment cannot however be considered inventive, as glutathione S transferase alpha serum level is already known to be inversely associated with cardiovascular risk as taught by D5. Therefore the subject-matter of claims 1, 2 and 5-7 is not inventive in the sense of Art. 56 EPC.

3.3 Alternatively taking document D5 as the closest prior art disclosing the measurement of plurality of serum biomarkers, including TNF alpha and GSTA1, in the context of cardiovascular risk, the objective technical problem to be solved in view of D5 could be seen as the provision of an alternative data analysis of the disclosed data.

Document D6 refers to Bayesian network modeling and exemplifies its use for epidemiologic system analysis of cardiovascular risk. The focus of data analysis is on the use of models able to discover and understand the relationships between different cardiovascular risk factors, see D6 the whole document, especially the abstract; figure 1; table 1.

Equally **document D1** refers to the establishment of a risk prediction model that can evaluate a patient with certain risk factors for the likelihood of the occurrence of coronary heart disease event within one year. The Risk Estimation Model and the Simplified Model were conducted using Bayesian networks and compared with Single Factor Models. D1 identifies a significant correlation between cardiovascular events and the biomarkers OPG, PIGF, Cathepsin S, GM-CSF, IP-10, CXCL-16, MIP1b, LDL-C, HDL-C, BMI, morphology of coronary lesions, creatine and high blood pressure, see D1 the whole document, especially the abstract and section "discussion" on pages 903-904.

The application of the Bayesian network modeling as disclosed in D6 and D1 for the measured data of D5 is considered obvious for the person skilled in the art rendering the subject-matter of claims 1, 2 and 5-7 not inventive, Art. 56 EPC.

3.4 Claim 7 defines the sample in the claimed method to comprise blood, urine or tissue. The application documents as filed do not however appear to provide any data for the biomarkers in urine or tissue samples. The application has thus not solved the problem over the whole claimed scope of claim 7, Art. 56 EPC.

4 Novelty and Inventive step / 2nd Invention searched (IL-6+GHR) - Art. 54 and 56 EPC

4.1 **Document D8** relates to the identification of biomarkers associated with subjects having an arteriovascular disease such as atherosclerosis, atherothrombosis, CAD, PAD, and CVD, are predisposed to or at risk for developing an arteriovascular disease or are predisposed to or at risk of experiencing an acute arteriovascular event. Such an arteriovascular disease is for example heart failure and such a cardiovascular event is for example myocardial infarction. The document refers to "arterio risk marker(s)" encompassing one or more of all biomarkers whose levels are changed in subjects who have an arteriovascular disease or are predisposed to developing an arteriovascular disease, or at risk of an arteriovascular event. Such "arterio risk markers" are summarized in Table 2 and include for example IL-6 (marker No 450), growth hormone receptor (GHR) (marker No 310) and TNF (marker No. 883). The document further teaches that an efficacious treatment is a treatment that leads to a decrease in the amount of a "arterio risk marker" protein. This implicitly discloses that an increased level of the "arterio risk marker" protein is indicative of an increased risk of suffering from a cardiovascular disease or a negative prognosis. Furthermore D8 refers to a kit containing detection reagents, oligonucleotides complementary to a portion of

the "arterio risk marker" nucleic acids, or antibodies to the proteins encoded by the "arterio risk marker" nucleic acids, packed together in the form of a kit, see D8 par. [0047]-[0050], [0056], [0057], [0097], [0239]-[0242]; table 2; and the claims.

Due to the disclosure of D8, the subject-matter of claims 1, 3, 5-8 is not novel, Art. 54 EPC.

Furthermore it is noted that the application of Bayesian network modeling as disclosed in **documents D6** and **D1** (see section 3.3 above) for the measured data of D8 is considered obvious for the person skilled in the art rendering the subject-matter of claims 1, 3 and 5-7 also not inventive in case of amending the claims to refer to Bayesian network modeling , Art. 56 EPC.

4.2 **Document D9** refers to evaluating the risk of major adverse cardiac event for subject by determining major adverse cardiac event risk score based on growth stimulation-expressed gene 2 (ST2), in combination with natriuretic peptide and comparing with reference. The disclosed method can also include measuring levels of other biomarkers in addition to ST2 and an NP, for example troponin, CRP, IL-6 and D-dimers, see D9 par. [0004]-[0007], [0049]. The document does not refer to the marker GHR.

Furthermore **document D4** refers to a study evaluating cardiovascular risk in patients with metabolic syndrome (MetS) and type 2 diabetes calculated using UKPDS software. D4 discloses the detection of plasma biomarkers including TNF alpha, and shows that increased plasma levels of pro-inflammatory cytokines TNF alpha and IL-6 are associated with increased cardiovascular risk, see D4 abstract "EAS-0943. The document does not refer to the marker GHR.

4.3 Claim 7 defines the sample in the claimed method to comprise blood, urine or tissue. The application documents as filed do not however appear to provide any data for the biomarkers in urine or tissue samples. The application has thus not solved the problem over the whole claimed scope of claim 7, Art. 56 EPC.

5 Novelty and Inventive step / 3rd Invention searched (RORA + TNF) - Art. 54 and 56 EPC

5.1 **Document D11** (EP2723893) refers to the measurement of mRNA expression of for example TNF-alpha and RORA using Affymetrix human U133A 2.0 array in the context of diagnosing rosacea, see D11 par. [0066]-[0070], figure 1. The disclosed kit can equally be considered suitable for the purpose as claimed in claim 8. The subject-matter of claim 8 is thus not novel over D11, Art. 54 EPC.

5.2 As already disclosed above **document D10** refers to the use of RORA gene in predicting risk of acute myocardial infarction (AMI) in patients with coronary heart disease. D10 discloses that the relative expression of RORA mRNA in peripheral blood of patients with AMI was significantly higher than that of SCAD (stable coronary artery disease) patients , see D10, page 1, section "technical field", section 2.3, and the claims.

Furthermore **document D12** identifies RORA as a relevant gene in the pathology of myocardial infarction (MI) and states that RORA was significantly down-regulated in MI compared to control, see D12 abstract; page 1458, left-column, 1st full paragraph; page 1463, left-column, 1st full par.; figure 7.

Also the marker TNF alpha is known in the context of risk of cardiovascular disease as disclosed by documents D3 and D4. **Document D3** discloses that patients with acute ischemic stroke compared to controls without stroke have significantly higher median serum levels of TNF-alpha. TNF-alpha levels were also positively associated with some established traditional cardiovascular risk factors, age, BMI, total cholesterol, blood pressure and inversely with HDL, see D3 page 9, left-column, 2nd par - right column, 2nd par.

Document D4 refers to a study evaluating cardiovascular risk in patients with metabolic syndrome (MetS) and type 2 diabetes calculated using UKPDS software. D4 discloses the detection of plasma biomarkers including TNF alpha, and shows that increased plasma levels of pro-inflammatory cytokines TNF alpha and IL-6 are associated with increased cardiovascular risk, see D4 abstract "EAS-0943".

5.3 Taking D10 or D12 as the closest prior art teaching the role of RORA in cardiovascular risk, the objective technical problem can be seen as the provision of an alternative/improved method of determining cardiovascular risk.

The application of the marker TNF alpha as an additional marker in the risk assessment cannot however be considered inventive, as increased blood TNF alpha levels is already known to be associated with cardiovascular risk as taught by D3 and D4.

Therefore the subject-matter of claims 1, 2 and 5-7 is not considered inventive in the sense of Art. 56 EPC over the teaching of D10 or D12 with D4, and the subject-matter of claims 1, 2, 6 and 7 cannot be considered inventive over D10 or D12 combined with D3, Art. 56 EPC.

5.4 Claim 7 defines the sample in the claimed method to comprise blood, urine or tissue. The application documents as filed do not however appear to provide any data for the biomarkers in urine or tissue samples. The application has thus not solved the problem over the whole claimed scope of claim 7, Art. 56 EPC.

6 Clarity - Art. 84 EPC / 1st, 2nd and 3rd invention searched

6.1 **Claim 1** is not considered clear as the claim fails to define how the expression level, amount, and/or activity of the biomarker is to deviate from the reference from a healthy control population (increased/decreased level, amount and/or activity) and its meaning (higher risk, lower risk), Art. 84 EPC.

6.2 The term "activity" used in **claim 1** is not clear, Art.84 EPC. According to the description of the application on page 3, the "*biomarker activity may relate to the activity of the biomarker protein, preferably activity in relation to interacting biomarkers*".

Furthermore it is noted that based on the description of the application, see pages 94-111, the applicants have applied computational analysis methods to analyse patient specific data (biomarker levels) in order to create a disease network, instead of "*detecting in a sample obtained from an individual the expression level/amount of the biomarker selected from a group..*" as claimed by claim 1. Thus the claim wording is considered not to reflect the method actually accomplished, Art. 84 EPC.

6.3 The wording of **claim 1** step c) and **claim 8** is not clear, as the claims refer to "*..prognosing the risk of an individual suffering from cardiovascular disease..*" without defining which risk is being prognosed, Art. 84 EPC. It is assumed that the claims should refer to "*..prognosing an individual's risk of suffering from cardiovascular disease..*", as used in the preamble of claim 1.